COMPUTER ENGINEERING
QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S4
DATE PERFORMED:	9/2/26	**DATE SUBMITTED:**	9/2/26
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ACTIVITY NO. 7.2 **Algorithm Analysis: Best-Case and Worst-Case Time Complexity**

1. PROCEDURE OUTPUT

Instructions: Analyze each algorithm carefully. For every item, determine the following:

1. **Best-Case Time Complexity**
2. **Worst-Case Time Complexity**
3. **Brief explanation** of how you arrived at your answers.

Use asymptotic notation such as **O(1), O(log n), O(n), O(n log n), O(n²)**, etc.

1. Finding an Element

```
None
for (int i = 0; i < n; i++)
{
    if (arr[i] == target)
    {
        return i;
    }
}
```

Best case: O(1)

Worst case:O(n)

Explanation: best target is first T(n) = 1 and the worst checks elements T(n) = n

2. Printing All Elements

```
None
for (int i = 0; i < n; i++)
{
```

```
    cout << arr[i] << endl;
}
```

Best case: O(n)

Worst case: O(n)

Explanation: Always prints n elements T(n) = n

3. Nested Loops

```
None
for (int i = 0; i < n; i++)
{
    for (int j = 0; j < n; j++)
    {
        cout << i << " " << j << endl;
    }
}
```

Best case: $O(n^2)$

Worst case: $O(n^2)$

Explanation: n x n = n^2 - $O(n^2)$

4. Conditional Processing

```
None
if (n > 100)
{
    for (int i = 0; i < n; i++)
    {
        cout << arr[i];
    }
}
else
{
    cout << "Small input";
}
```

Best case: O(1)

Worst case: O(n)

Explanation: if ≤ 10o, only the else executes O(1). If n > 100, loop executes n times O(n)

5. Fixed Inner Loop

```
None
for (int i = 0; i < n; i++)
{
    for (int j = 0; j < 10; j++)
    {
        cout << i << j;
    }
}
```

Best case: O(n)

Worst case: O(n)

Explanation: n x 10 = 10n - O(n)

6. Two Consecutive Loops

```
None
for (int i = 0; i < n; i++)
{
    cout << arr[i];
}

for (int j = 0; j < n; j++)
{
    cout << arr[j] * 2;
}
```

Best case: O(n)

Worst case: O(n)

Explanation: n + n = 2n - O(n)

7. Dividing by Two

None
```
int x = n;

while (x > 1)
{
    x = x / 2;
}
```

Best case: O(log n)

Worst case: O(log n)

Explanation: x is divided by 2 - log n iterations - o(log n)

8. Nested Loop With Different Limits

None
```
for (int i = 0; i < n; i++)
{
    for (int j = 0; j < i; j++)
    {
        cout << i << j;
    }
}
```

Best case: O(n^2)

Worst case: O(n^2)

Explanation: Work is 0+1+2+...+(n-1) = n(n-1)/2 → O(n^2).

9. Early Termination

None
```
for (int i = 0; i < n; i++)
{
    if (arr[i] == target)
    {
        cout << "Found";
        break;
    }
```

```
}
```

Best case: O(1)

Worst case: O(1)

Explanation: Best: found at first position. Worst: checks all n elements.

10. Two-Level Search

```
None
for (int i = 0; i < n; i++)
{
    for (int j = 0; j < n; j++)
    {
        if (arr[i] == target)
        {
            break;
        }
    }
}
```

Best case:

Worst case:

Explanation:

11. Linear Loop With Logarithmic Inner Work

```
None
for (int i = 0; i < n; i++)
{
    int x = n;

    while (x > 1)
    {
        x = x / 2;
    }
}
```

Best case:

Worst case:

Explanation:

12. Conditional Nested Loop

```
None
if (arr[0] == target)
{
    cout << "Found";
}
else
{
    for (int i = 0; i < n; i++)
    {
        for (int j = 0; j < n; j++)
        {
            cout << arr[i] + arr[j];
        }
    }
}
```

Best case: O(1)

Worst case: $O(n^2)$

Explanation: Best: arr[0] == target, so nested loops aren't executed. Worst: nested loops → n × n = n^2.

13. Finding the Maximum

```
None
int max = arr[0];

for (int i = 1; i < n; i++)
{
    if (arr[i] > max)
    {
        max = arr[i];
    }
```

```
}
```

Best case: O(n)

Worst case: O(n)

Explanation: Must examine all elements → T(n)=n-1 → O(n)

14. Three Nested Loops

```
None
 for (int i = 0; i < n; i++)
 {
     for (int j = 0; j < n; j++)
     {
         for (int k = 0; k < n; k++)
         {
             cout << i + j + k;
         }
     }
}
```

Best case:

Worst case:

Explanation:

15. Comprehensive Analysis

```
None
for (int i = 0; i < n; i++)
{
    if (arr[i] == target)
    {
       return i;
    }
    int x = n;

    while (x > 1)
```

```
    {
        x = x / 2;
    }

}
```

Best case: O(1)

Worst case: O(n log n)

Explanation: Best: target is first → returns immediately. Worst: n outer iterations × log n inner work → O(n log n).

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

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Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

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